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C–H Activation

Cobalt-Catalyzed C8-Dienylation of Quinoline-*N*-Oxides

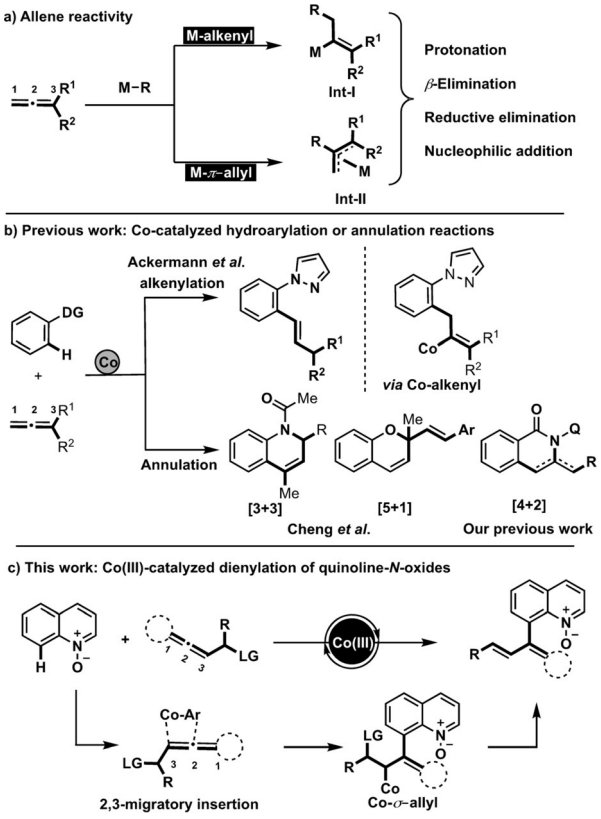
*Rahul K. Shukla, Akshay M. Nair, Salman Khan, and Chandra M. R. Volla**

Abstract: *An efficient Cp*Co^{III}-catalyzed C8-dienylation of quinoline-N-oxides was achieved by employing allenes bearing leaving groups at the α-position as the dienylating agents. The reaction proceeds by Co^{III}-catalyzed C–H activation of quinoline-N-oxides and regioselective migratory insertion of the allene followed by a β-oxy elimination, leading to overall dienylation. Site-selective C–H activation was achieved with excellent selectivity under mild reaction conditions, and 30 mol% of a NaF additive was found to be crucial for the efficient dienylation. The methodology features high stereoselectivity, mild reaction conditions, and good functional-group tolerance. C8-alkenylation of quinoline-N-oxides was achieved in the case of allenes devoid of leaving groups as coupling partners. Furthermore, gram-scale preparation and preliminary mechanistic experiments were carried out to gain insights into the reaction mechanism.*

Introduction

Allenes have emerged as valuable scaffolds in synthetic organic chemistry and have been widely studied in a variety of carbocyclization and cross-coupling reactions.[1] The recent years have witnessed a remarkable surge in the study of allenes as coupling partners in transition metal catalyzed C–H activation reactions.[2] The presence of two orthogonal double bonds in an allene moiety gives rise to a variety of regioselective and chemoselective disparities, which impart significant challenges for engaging allenes in transition metal mediated reactions, and at the same time provide ample opportunities to generate rapid molecular complexity by exploiting their unique reactivity patterns.[3] For example, depending on the stereoelectronic nature of the allene and the choice of the metal catalyst **M**, the organometallic intermediate **R-M** generated after C–H activation, could follow two distinct migratory insertion pathways (Scheme 1a). The 1,2-insertion of **R-M** on the allene would lead to the metal alkenyl intermediate **Int-I**, while the carbometallation with the C2=C3 double bond of allenes leads to the π-allyl intermediate **Int-II**.[4] Both these intermediates are highly reactive and undergo a variety of termination steps such as protonation, β-hydride elimination, reductive elimination, or Tsuji—Trost-type nucleophilic addition resulting in an assortment of products. Capitalizing on these complementary reactivities, a variety of C–H functionalization reactions like alkenylation, allenylation, and other reactions have been reported, employing precious second- and third-row transition-metal catalysts like Ir^{III}, Rh^{III}, Ru^{II}, and Pd^{II}.[5–8]

The quest for efficient and sustainable catalysis, has led to the emergence of low-cost and environmentally benign first-row transition metals as worthy catalysts in C–H functionalizations.[9,10] Amidst the renaissance in earth-abundant metal catalysis, cobalt has emerged as an evident choice for the development of C–H functionalization protocols because of its higher Lewis acidity and good selectivity.[11] A variety of C–H functionalization reactions using both low-valent and high-valent cobalt catalysts have been reported in the recent years. However, the relevance of a cobalt catalysis for C–H functionalization with allenes has been infrequently reported and the reactivity is so far limited to alkenylation and annulation reactions (Scheme 1b).[12] Ackermann and co-workers illustrated an elegant C–H alkenylation with allenes originating from cobalt alkenyl pathway (**Int-I**).[13] Later, the complementary cobalt-allyl intermediate generated from allenes was explored by Cheng and co-workers to describe



Scheme 1. Overview of the work.

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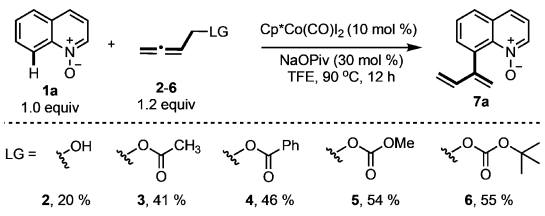
[3+3] and [5+1] annulation reactions.[14] In this regard, we recently reported a cobalt catalyzed [4+2] annulation reaction of allenes and benzamides for the regioselective synthesis of dihydroiso quinolinones.[15] We observed that the reaction proceeds via a Co-σ-allyl complex formed by the regioselective migratory insertion of the arylcobalt(III)-intermediate with the C2=C3 double bond of allenes. To gain more understanding on this mechanistic rationale and to expand the chemical space, we envisioned installing a leaving group at the allylic position of the allene for allowing a facile β-oxy elimination to realize a catalytic dienylation under oxidant-free conditions. While the transition metal catalyzed C–H allylation reactions have been extensively studied, the corresponding C–H dienylation reactions to access functionalized 1,3-dienes were largely ignored, despite their high relevance in organic synthesis and materials science.[16] The traditional methods for accessing dienes involve Wittig/HWE reactions or transition metal mediated cross-coupling reactions.[17] However these reactions often require prefunctionalized substrates and suffer from poor stereo- and regioselectivity. Consequently, the development of suitable dienylating agents for regioselective introduction of diene fragment into organic molecules is of utmost importance.[18]

At the same time, functionalization of quinoline at various positions is of prime interest to both medicinal and synthetic chemists alike.[19] The site-selective transformations of quinoline at the C2(H) position is well explored because of its acidic nature.[20] However, only a handful of reports exist documenting C8(H) functionalization of quinolines.[21] Selective C8 functionalization of quinoline-*N*-oxides using transition-metal catalysts has gained significant attention in the recent years. Owing to the importance of both functionalized 1,3-dienes and quinolines in medicinal chemistry, we envisioned developing a site-selective catalytic dienylation of quinoline-*N*-oxides using allenylcarbinol carbonates as the dienylating agents. Based on our previous work, we anticipated that the Co-σ-allyl complex formed after regioselective 2,3-migratory insertion will undergo facile β-oxy elimination, leading to C2-functionalized dienes (Scheme 1c).

Results and Discussion

Identification of the Leaving Group

To facilitate a facile β-oxy elimination from a Co^{III}-σ-allyl-intermediate, the allenyl alcohol **2** was chosen as the dienlylating agent (Scheme 2). To our delight, the C8-dienylated quinoline-*N*-oxide **7a** was afforded in 20% when



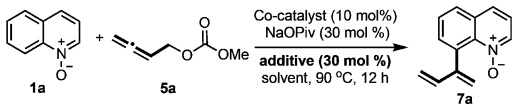
Scheme 2. Optimization of the leaving group.

1a was reacted with **2** using 10 mol% of Cp*Co(CO)I_2 and 30 mol% of NaOPiv in TFE at 90 °C for 12 hours. Delighted with this, we employed the allenyl acetate **3** instead of **2** in reaction with **1a** and **7a** was obtained in an improved yield of 41%. This motivated us to further screen various leaving groups on the allene (Scheme 2) and among these leaving groups, as expected, carbonate was found to be most favorable for the present transformation as allenyl carbinol carbonate **5** delivered **7a** in 54% and the *tert*-butyl carbonate (Boc) **6** led to the product in 55%.

Optimization of the Reaction Conditions

To achieve further amplification in yield, a solvent screening was carried out and we observed that the reaction proceeds more proficiently in TFE compared to other solvents like MeOH, DCE, HFIP (Table 1, entries 1–6). Inquisitively, when the effect of various bases was studied, we found that NaOPiv was optimal for the dienylation (entries 7 and 8). Further, we investigated the consequence of ionization by the addition of silver salts. Addition of 30 mol% of AgOTf furnished the dienylation product **7a** in 60% (entry 10) and 30 mol% of $AgSbF_6$ led to the product in 55% (entry 11). Interestingly, a substantial amplification in yield was observed when KF (78%) was used as an additive (entry 12) and captivatingly, the yield increased to 88% (82% upon isolation) when NaF was employed (entry 13). With

Table 1: Optimization of reaction conditions.[a]



Entry	Co-catalyst	Additive	Solvent	Yield [%][b]
1	Cp*Co(CO)I_2	–	TFE	54 (49)[c]
2	Cp*Co(CO)I_2	–	MeOH	10
3	Cp*Co(CO)I_2	–	*t*-Am-OH	18
4	Cp*Co(CO)I_2	–	CH_3NO_2	n.r.
5	Cp*Co(CO)I_2	–	DCE	n.r.
6	Cp*Co(CO)I_2	–	HFIP	20
7[d]	Cp*Co(CO)I_2	–	TFE	29
8[e]	Cp*Co(CO)I_2	–	TFE	41
9	$(Cp^*CoI_2)_2$	–	TFE	48
10	Cp*Co(CO)I_2	AgOTf	TFE	60
11	Cp*Co(CO)I_2	$AgSbF_6$	TFE	55
12	Cp*Co(CO)I_2	KF	TFE	78
13	Cp*Co(CO)I_2	NaF	TFE	88 (82)[c]
14	Cp*Co(CO)I_2	AgF	TFE	86
15	Cp*Co(CO)I_2	NaI	TFE	53
16	Cp*Co(CO)I_2	NaBr	TFE	56
17	Cp*Co(CO)I_2	NaCl	TFE	65
18	Cp*Co(CO)I_2	*n*-Bu_4NBF_4	TFE	80
19	$[Ru(p\text{-cymene})Cl_2]_2$	NaF	TFE	n.r.

Reaction conditions: [a] 0.2 mmol of quinoline-*N*-oxide **1a**, 0.24 mmol **5a**, 10 mol% Co-catalyst, 30 mol% NaOPiv in 1 mL solvent. [b] NMR yield was taken using 0.1 mmol of 1,3,5-trimethoxybenzene as an internal standard. [c] Yield within parentheses refers to the yield of the isolated product. [d] 30 mol% of CsOAc instead of NaOPiv. [e] 30 mol% of NaOAc instead of NaOPiv.

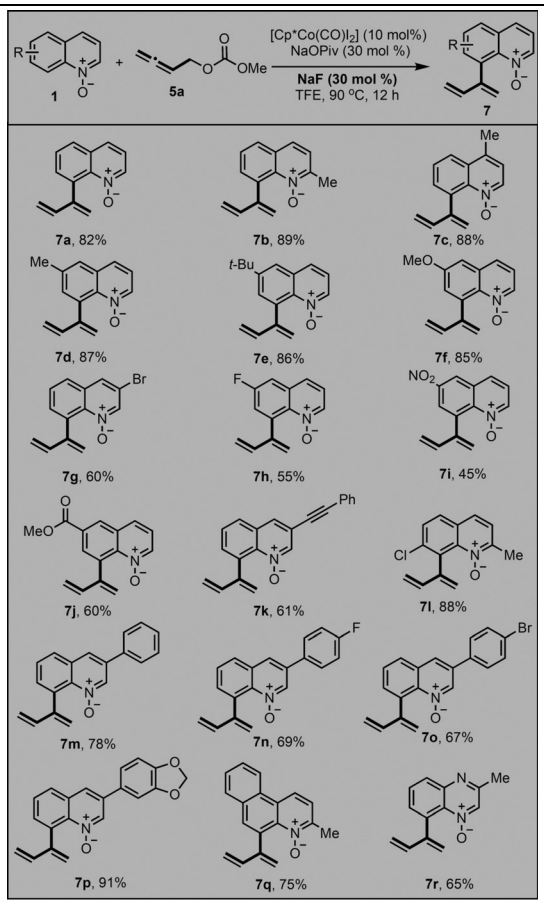
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further screening of various additives (entries 14–18), we found the optimized reaction conditions to be the following: 1 equivalent of **1a** and 1.2 equivalents of **5a** in the presence of 10 mol% of $Cp^*Co(CO)I_2$, 30 mol% of NaOPiv, and 30 mol% of NaF in TFE at 90 °C affords **7a** in 82%.

With the optimized reaction conditions in hand, we investigated the scope of various diversely substituted quinoline-*N*-oxides (Table 2), which could participate in the dienylation reaction. Alkyl groups on various positions of the *N*-oxide were well tolerated, giving excellent yields of the corresponding products (**7b—e**, 82–89%). While electron-rich quinoline-*N*-oxides having alkyl and methoxy groups provided the corresponding products in good yields, moderate yields were observed for the electron-deficient counterparts (**7g–j** in 45–60%). To our delight, quinoline-*N*-oxides bearing precarious functional groups such as nitro (**1i**), ester (**1j**), and alkyne (**1k**) furnished the respective products **7i–k**, albeit in slightly diminished yields. Moreover, the disubstituted quinoline *N*-oxide **1l** was also found to be an amenable substrate providing **7l** in 88%. Quinoline-*N*-oxides bearing an array of electronically diverse 3-aryl substituents furnished the corresponding products **7m–q** in good yields (67–91%). Remarkably, the reaction can be extended for C5- and C8-dienylations of the benzo[*f*]quinoline-*N*-oxide **1q** and quinoxaline *N*-oxide **1r**, respectively, affording the corresponding products **7q** (75%) and **7r** (65%). However, tetrahydroquinoline-*N*-oxide (**7s**) and 2-vinyl-pyridine-*N*-oxide (**7t**) were found to be ineffective indicating that the dienylation is selective for quinoline-*N*-oxides.

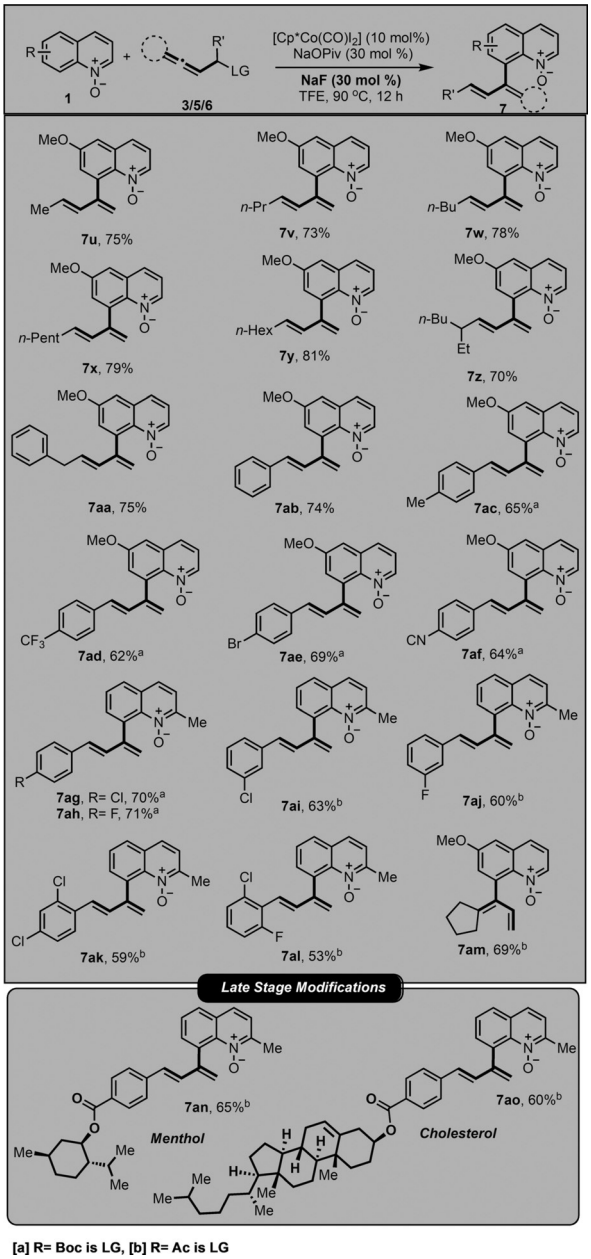
Next the scope of the developed protocol was studied against an array of allenyl carbinol carbonates (Table 3). Allenes bearing alkyl chains of increasing chain length (R = methyl, *n*-propyl, *n*-butyl, *n*-pentyl and *n*-hexyl) fared well for the reaction furnishing the corresponding dienes **7u–y** in 73–81%. While the branched allene furnished the product **7z** in

Table 2: Scope of the reaction with various quinoline *N*-oxides.



Reaction conditions: **1** (0.2 mmol), **5a** (0.24 mmol), [$Cp^*Co(CO)I_2$] (10 mol%), NaOPiv (30 mol%), NaF (30 mol%) in 1 mL TFE at 90 °C.

Table 3: Scope of the reaction with different allenyl carbinol carbonate.



[a] R= Boc is LG, [b] R= Ac is LG

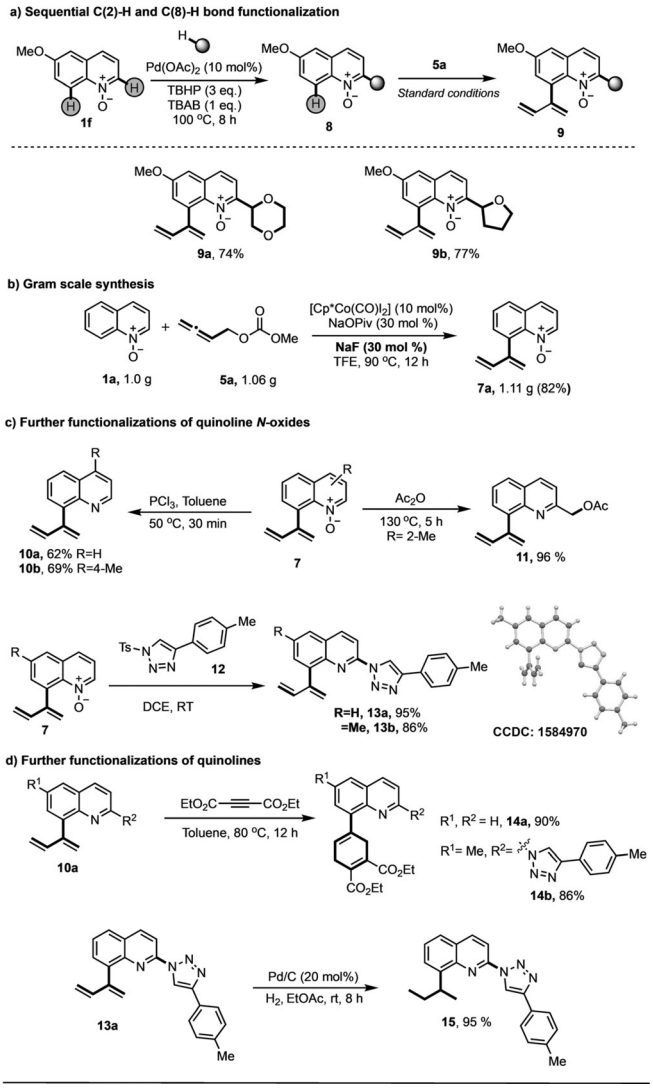
Reaction conditions: **1** (0.2 mmol), **5** (LG = OCO_2Me, 0.24 mmol), [$Cp^*Co(CO)I_2$] (10 mol%), NaOPiv (30 mol%), NaF (30 mol%) in 1 mL TFE at 90 °C. Yields of isolated products provided. [a] **6** (LG = OBoc) is used instead of **5**. [b] **3** (LG = OAc) is used instead of **5**.

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70% yield, **7aa** was obtained in 75% when benzyl-substituted allene was employed. Allenes bearing electronically diverse phenyl rings were also found to be amenable furnishing the corresponding 1,3-diarylated 1,3-dienes **7ab–al** in 53–74% yields. The reaction was also extended to internal allenes to isolate the corresponding diene **7am** in 69%. Further, we found the protocol was amenable for late-stage modifications of complex natural products. Allenes tethered to menthol and cholesterol furnished the corresponding dienes **7an** (65%) and **7ao** (60%) in good yields.

To epitomize the utility of the protocol, we envisioned a sequential hetero-bisfunctionalization of quinoline-*N*-oxide by carrying out a Pd^{II}-catalyzed C2-alkylation followed by Co^{III}-catalyzed C8-dienylation. Pd-catalyzed C2-alkylation of 6-methoxy quinoline-*N*-oxide (**1f**) was considered based on the work of Wu and co-workers[22] with 1,4-dioxane and tetrahydrofuran to access the corresponding C2-functionalized quinoline-*N*-oxides **8a** and **8b**, respectively. Co^{III}-catalyzed dienylation of **8** with **5a** led to the smooth formation of C2- and C8-functionalized 6-methoxyquinoline *N*-oxides **9a** and **9b**, respectively, in good yields (Scheme 3a). A gram-scale reaction was performed to probe the synthetic utility of the developed reaction. When 1.0 gram of **1a** was reacted with 1.06 grams of **5a** under the standard reaction conditions, 1.11 grams of **7a** were obtained in 82% yield, showing the practicality of the dienylation (Scheme 3b). Quinoline *N*-oxide acts as a traceless directing group that allows further diversification at late stages by keeping the butadiene fragment intact as depicted in Scheme 3c. Reduction of the quinoline *N*-oxides to synthetically more relevant quinolines was achieved on treatment with PCl_3 to isolate **10a** and **10b** in 62% and 69%, respectively. In addition, reaction of **7b** with acetic anhydride led to the clean transposition of the oxy functionality to the α-carbon atom, affording the α-acetoxy-substituted quinaldine **11** in 96% yield. Given the importance of both quinolines and triazoles in medicinal chemistry, it could be expected that their conjugates could exhibit significant pharmacological activities. With this in mind, we explored the reactivity of C8-dienylated quinoline *N*-oxides with *N*-sulfonylated 1,2,3-triazoles and we were delighted to observe the formation of the 2-triazolyl quinolines **13a** (95%) and **13b** (86%) in excellent yields by treating **7** with triazole **12**.[23] The structure and regioselectivity were unambiguously confirmed by the single-crystal X-ray diffraction analysis of **13b**. Further, an intermolecular Diels–Alder reaction of 1,3-diene **10a** with diethyl acetylene dicarboxylate proceeded efficiently in toluene at 80 °C to afford cyclic 1,4-diene **14a** in 90% yield. Additionally, catalytic hydrogenation of **13a** using 20 mol% of Pd/C provided the formal alkylation product **15** in 95% yield.

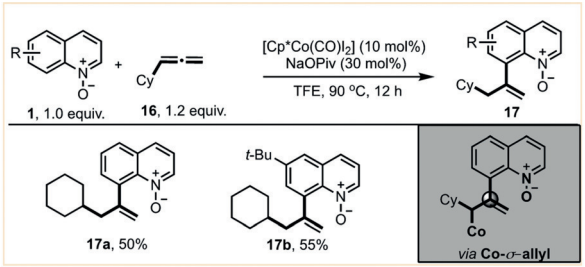
Subsequently, to gain insights on the reaction mechanism and to expand the synthetic applicability, we tested the reactivity of allenes, devoid of a leaving group at the β-position, for the Co^{III}-catalyzed C–H activation of quinoline *N*-oxides (**1**). We were pleased to observe that treating **1** with the cyclohexyl allene **16** using 10 mol% $[Cp^*Co(CO)I_2]$ in TFE led to the formation of the C8-alkenylated quinoline *N*-oxides **17a**,**b** in moderate yields (50–55%; Scheme 4). This result could be attributed to the protonation of the cobalt-σ-allyl intermediate originating from the 2,3-insertion on to the allene. The formation of both **7** and **17** suggests the tendency of arylcobalt(III)-intermediate to undergo regioselective 2,3-migratory insertion with allenes.



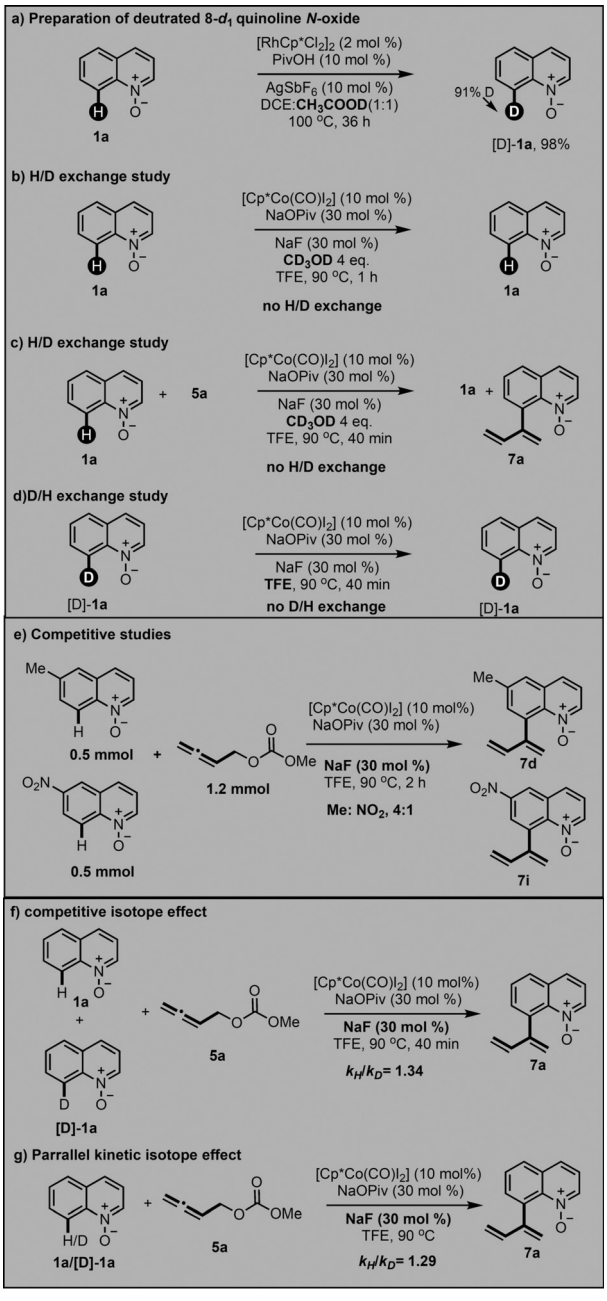
Scheme 3. Sequential functionalization, gram-scale and further functionalizations. The X-ray structure is that of **13b**.[24]

To shed light on to the reaction mechanism, a series of preliminary mechanistic studies were performed. The C8-deuterated quinoline *N*-oxide [D]-**1a** (91% D) was synthesized in 98% by Rh^{III}-catalyzed C–H activation of **1a** in the presence of 1:1 mixture of DCE and CH_3COOD (Scheme 5a). Deuterium-exchange studies were carried out under the standard reaction conditions with **1a** and [D]-**1a** and no deuterium incorporation was detected at the C8-position, with or without **5a**, suggesting that the C–H bond cleavage (C–H activation) is irreversible (Scheme 5b–d). Further, competitive experiments with 6-Me- and 6-NO_2-substituted quinoline-*N*-oxides with **5a** indicated a preference for the electron-rich 6-Me derivative, probably because of its better coordinating ability compared to 6-NO_2 derivative (Scheme 5e). A competitive isotopic value (k_H/k_D) of 1.34 was

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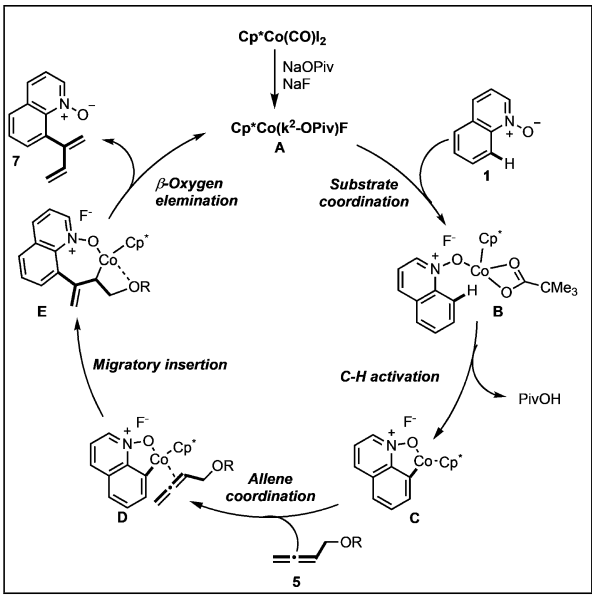
Scheme 4. Co^{III}-catalyzed C8-alkenylation of quinoline-*N*-oxides.



Scheme 5. Mechanistic studies.

observed when a 1:1 mixture of **1a** and C8-[D]-**1a** was reacted with **5a** under the standard reaction conditions (Scheme 5 f). Finally, parallel reactions of **1a** or C8-[D]-**1a** with **5a** showed a kinetic isotopic value (k_H/k_D) of 1.29 (Scheme 5 g). These values imply that C–H bond cleavage might not be involved in the rate-limiting step.

Based on our mechanistic studies, the plausible mechanism for the Co^{III}-catalyzed dienylation is proposed as shown in Scheme 6. Ionization of $Cp^*Co(CO)I_2$ in the presence of sodium carboxylate and sodium fluoride forms the catalytically active Co^{III}-complex **A**. Coordination of **1** with **A** proceeds through electron-rich oxygen, followed by site-selective C–H activation at the C8-position, leading to the cyclometalated complex **C**. Coordination of **C** with allenyl carbinol carbonate **5** followed by regioselective migratory insertion with C2=C3 double bond leads to the intermediate **E**. Finally, facile β-oxy elimination from **E** furnishes the dienylated product **7**, leading to overall dienylation.



Scheme 6. Proposed reaction mechanism.

Conclusion

In conclusion, we have developed the first cobalt-catalyzed regioselective C8 dienylation of quinoline *N*-oxides, employing allenylcarbinol carbonates as the dienylating agents. The reaction displays broad substrate scope with good to excellent yields against a variety of diversely substituted quinoline *N*-oxides and allenes under relatively mild reaction conditions. The utility of the protocol was justified by the facile scalability and further synthetic transformation of the products. Furthermore, to expand the generality of the protocol, 1,2,3-triazole conjugates of the dienylated quinolines were synthesized in excellent yields. Competitive experiments and deuterium-labeling studies were carried out shed light on the reaction mechanism.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: allenes · C–H activation · cobalt · heterocycles · N-oxides

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[24] CCDC 1584970 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre.

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